

Sivan Syed

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Austin, TX

🌐 <https://ilgneous.github.io/>

Education

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

B.S. Electrical Engineering, B.S. Physics, B.S. Mathematics, Certificate German

Relevant Coursework: Fluid Dynamics, Mechanics of Solids, Materials Engineering, Thermodynamics, Dynamics

Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics I

Experience

Emerson, Electrical Engineering Design Intern

May 2026–Aug 2026

- Authored configuration and assembly drawings from scratch for the LT-1 transmitter, documenting every valid mounting and assembly variant
- Reverse-engineered a raw production PCB to author complete schematics and assembly Work Instructions
- Wrote assembly Work Instructions by auditing the manufacturing floor, tying engineering documentation directly to production

Adom Industries Inc., R&D Design Engineer

May 2025–Aug 2025

- Optimized chassis topology through iterative FEA, cutting material usage 88% while preserving structural integrity for payload transfer
- Architected a 48V VCU (RP5+RP2040) for ROS path planning via CAN and a hardware E-stop and isolated 5V logic rail
- Authored Design for Manufacturing (DFM) docs covering wiring looms, thermal interfaces, and EMI shielding

Sandia National Laboratories, R&D Computational Engineer

May 2024–Aug 2024

- Modeled radial Beryllium liner implosions with ODE/trajectory solvers, comparing ALEGRA predictions against observed fluid instability spectra (RTIs/RMIs)

Banerjee Lab – 2D Material Synthesis & Characterization, Undergraduate Researcher

Sep 2025–Present

- Characterized and grew MoS₂ monolayers via CVD to optimize carrier type, doping, and bandgap
- Employed XRD, Raman, and Photoluminescence spectroscopy for material analysis

FSAE Longhorn Racing, Dynamics + Vehicle Modeling Engineer, Steering Lead

Sep 2022–Aug 2025

- Re-engineered the steering column from steel to a carbon-fiber layup with a metal 3D-printed bevel gearbox, cutting weight 68% at equal torsional rigidity
- Tuned anti-Ackermann geometry, roll center, and bump steer in a Model-in-the-Loop simulation, validating tire and slip-angle targets against track telemetry

Projects

Racing Simulator – Pedal Box & Steering Assembly

- Designed a fully constrained spring-actuated pedal box with driver-adjustable travel and a coaxial rotary-encoder mount for displacement sensing
- Modeled a table-mounted gyroscope-integrated steering assembly for driver use and PCB mounting

UAV Airframe & Flight Controller

- Modeled the rigid-body flight dynamics about hover to size motor/propeller actuation and shape the closed-loop attitude response of a custom quadrotor airframe
- Designing a deterministic 400 Hz attitude-control loop on a custom flight controller with fault-tolerant motor handling, hardening the airframe electronics against ESC vibration and EMI

High-Power E-Bike Boost Converter

- Engineered a custom DC-DC boost converter PCB, suppressing radiated EMI by minimizing the high di/dt critical loop and gate-drive parasitic inductance

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Tools: KiCAD, Cadence, Draftsight, LTSpice, MuMax3, Fusion360, SOLIDWORKS, Ansys